

Retrieval-Augmented Generation (RAG) is an architectural approach that improves the efficacy of large language models (LLMs) by grounding them in external knowledge bases. Traditional LLMs rely solely on their parametric memory—the knowledge embedded in their weights during pre-training. This often leads to "hallucinations," where the model generates plausible but factually incorrect information when asked about proprietary or post-training data.

A standard RAG pipeline consists of two primary components: the Retriever and the Generator. The Retriever converts user queries into dense vector embeddings using models like Sentence Transformers, then performs a similarity search against a vector database containing chunked documents. The top-K most relevant chunks are retrieved and appended to the user's original prompt. The Generator, typically a sequence-to-sequence model like BART or a decoder-only model like LLaMA, then synthesizes a final response using both the original query and the retrieved context. This ensures the output is verifiable and up-to-date without requiring expensive model fine-tuning.